

Managers' Understanding of Artificial Intelligence in Relation to Marketing Financial Services: Insights from a Cross-Country Study

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4 **Managers’ Understanding of Artificial Intelligence in Relation to**
5 **Marketing Financial Services: Insights from a Cross-Country Study**
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8 **Abstract**
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10 **Purpose** – Given that managers play a crucial role in developing and deploying AI
11 for marketing financial services, this study was aimed at better understanding their
12 awareness regarding AI and the challenges they are facing in providing the
13 attendant technologies, as well as highlighting key stakeholders and their
14 collaborative efforts in providing financial services.
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18 **Design/methodology/approach** – Exploratory, inductive research design. The
19 data was gathered through semi-structured interviews with 47 bank managers in
20 both developed and developing countries, including the United Kingdom, Canada,
21 Nigeria and Vietnam.
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25 **Findings** – Managers are aware of the prospects of AI and are making efforts to
26 address AI as a business need but find that there often exist certain challenges in
27 accelerating AI adoption. The study also presents a conceptual framework of AI in
28 relation to financial service marketing, which captures and highlights the
29 interactions among the customers, banks, and external stakeholders, as well as the
30 regulators.
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35 **Research limitations/implications** – Banks must understand their business
36 objectives, the available resources, and the needs of their customers. Managers
37 should keep the ethical implications of their working relationships in mind when
38 selecting a team or collaborating with partners. In addition, managers should be
39 trained and assisted in comprehending AI in relation to financial services, while
40 the regulators must be involved in the development of AI for financial service
41 marketing. Finally, it is critical to communicate the prospects for AI to consumers.
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46 **Originality/value** – This study provides empirical insight into the opportunities,
47 prospects and challenges pertaining to the use of AI in the area of financial service
48 marketing. It also specifically calls into question certain preconceptions regarding
49 AI and its role in financial services, the chatbots adopted for financial service
50 delivery, and the role of marketing managers in developing AI.
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55 **Keywords:** Managers, Artificial Intelligence, Marketing, Qualitative, Financial
56 Services, Banking
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58 **Paper type:** Research paper
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Introduction

The introduction of artificial intelligence (AI) in the financial services industry largely arose from the increasing demand for financial regulation, the need for profitability, and the competition among firms (Akyüz and Mavnacıoğlu, 2021). Meanwhile, it is also associated with the advancements in data resources and the associated technologies, as well as the availability of the necessary financial sector infrastructure. The field of AI largely involves the adoption of intense computer science applications and is grounded in the fields of psychology, linguistics, mathematics, and philosophy (Fernandez, 2019). It has been established as a powerful tool in the provision of financial services, and many firms are currently utilising various AI-based analytical tools, including machine learning for data analysis.

Recent research has demonstrated that the number of organisations adopting AI in their business practises has increased by 270% in the last four years, with nine out of ten leading businesses making ongoing investments in AI, meaning the global AI market is expected to reach \$267 billion by 2027 (Lin, 2020). According to Insider Intelligence's report on the use of AI in the banking industry, most banks (80%) are aware of the potential benefits presented by AI, and many have implemented AI for risk management or revenue generation (Digalaki, 2021). However, most companies remain in the learning phase of adopting AI, continuing to explore how best to integrate various AI systems into their business operations (Liu, 2020), and this has called for a better theoretical understanding of the scope of AI and how managers can adopt AI for their business operations.

While various studies have explored AI in relation to business operations in terms of marketing and customer engagement (e.g., Davenport et al., 2020; Dwiledei et al., 2020;

Norbert, 2018), there is a growing amount of research on how AI is specifically being adopted for marketing financial services and how it is enhancing the attendant consumer experience. For example, Riikkinen et al. (2018) explored the use of chatbots in the insurance industry, while both Abduladri et al. (2021) and Mogaji et al. (2021) examined chatbots as a form of digital transformation in the provision of financial services. Elsewhere, Mogaji et al. (2021) explored the adoption of AI in marketing financial services in terms of vulnerable customers, while Jang et al. (2021) investigated the understanding of chatbots among the managers in the Korean financial industry.

Despite this growing body of work, there remain gaps in the knowledge on the use of AI in the area of financial services marketing. First, the existing studies have predominantly focused on chatbots as the main feature of AI in this field (e.g., Abduladri et al., 2021; Mogaji et al., 2021; Riikkinen et al., 2018), often ignoring the other technological innovations AI has provided, such as credit evaluation tools, credit score evaluation, and bankruptcy prediction (Donepudi, 2017). Second, the previous studies have often focused on a specific country without holding a holistic view of the digital transformation ensuing throughout the industry. For example, Jang et al. (2021) focused on Korean bank managers, Eren (2021) on customers in Turkey, and Mogaji et al. (2021) on Nigerian customers. Third, prior research has frequently concentrated on consumers’ adoption and experiences of using AI systems, while, with the exception of Jang et al. (2021), researchers have largely ignored the managers’ perspectives on the adoption of AI.

The above present the gaps in knowledge that this study aims to fill through collecting and analysing data collected from managers operating in various different countries –

from both the developed and the developing worlds – and across different sectors within the financial services industry.

Given that the managers play a crucial role in developing and deploying the AI systems and ensuring that the customers adopt these technologies (Jang et al., 2021), it is essential to understand the perspective and challenges they are facing in providing these technologies. Therefore, this study makes a significant theoretical contribution to our understanding of AI in relation to financial studies, extending the previous studies, such as those from Mogaji et al. (2020), Riikinen et al. (2018), and Jang et al. (2021), by exploring AI from a different perspective. Specifically, the study presents a conceptual framework for AI in relation to financial service marketing research, highlighting key stakeholders and their working relationships in providing customers with enhanced financial services (see Table 1 for summary of key studies and contribution of our study). Furthermore, the study provides various managerial implications for financial service providers and Fintech developers in terms of the prospects, opportunities, and challenges pertaining to the adoption of AI for financial service marketing.

Literature review

The application of artificial intelligence for marketing

While it is widely recognised that AI is applicable and being adopted in terms of numerous facets of human endeavour, this paper focuses on adopting AI for marketing and consumer engagement within the financial services industry. To date, AI has dominated a number of technology-centred segments, and its influence will soon manifest in of marketing. While Norbert (2018) chose to question the feasibility of developing AI-based marketing insights, a number of recent studies have highlighted the enormous

prospects of AI in terms of improving marketing strategies. For example, Dwiledei et al. (2020) presented a seminal work exploring the significant opportunities and prospects of AI in marketing and other business practices, with the authors acknowledging that AI will shape the future of the industry and society at large. This notion was further corroborated by Davenport et al. (2020), who postulated how AI will change the future of marketing, citing the algorithms' capacity to automate business processes, analyse Big Data, and gain insight from both customer and transaction data.

The integration of AI in the area of digital marketing was largely influenced by the advances in computing power and the emergence of Big Data (Haenlein and Kaplan, 2019). With various AI tools, organisations can better understand the situation and subsequently implement effective target marketing strategies (Mogaji et al., 2020). Meanwhile, the onset of digital technologies has compelled marketers to reconsider the traditional marketing techniques, with digital marketing regarded as being a more viable alternative (Saura et al., 2019). According to Mogaji et al. (2020), AI provides viable tools for analysing the precarious conditions of vulnerable customers and the suitable contact methods for implementing digital marketing for this segment.

The application of AI and machine learning in marketing allows for capitalising on the capacity for establishing a connection between computing power and the vast human insights. According to Ma and Sun (2020), AI agents powered by machine learning are continuously transforming the business world. This transformation largely relies on the combined capacities of AI and machine learning, which are used in the areas of marketing research, data analysis, tracking, and networking, to establish consumer purchase patterns and connect them to specific marketing theories. For their part, Shahid and Li (2019)

established that the application of AI in marketing helps to improve business performance through its deployment in terms of various marketing functions. In fact, AI is being used to develop numerous marketing strategies and other marketing-related activities, such as product promotion, pricing, planning, development, and distribution (Mogaji et al., 2020).

The application of artificial intelligence in financial services

The major application of AI has been the provision of various financial services, with its deployment in this field allows for the capitalisation on increasing possibilities of AI that have developed over recent years (Fernandez, 2019). In fact, the attendant advancements have made it possible to develop numerous practical applications that serve various needs in the financial services industry, including the automation of various tasks and the expansion of the analytical capacity. Here, the efficiency is increased, the quality is enhanced, customer satisfaction is improved, and the operation costs are reduced (Grigoroudis et al., 2012). Meanwhile, repetitive tasks can be easily automated using AI, with such an intervention minimising human error while boosting productivity. The technology is also deployed in the financial services industry to enhance analytical capacities since it enables financial service providers to analyse larger volumes of both structured and unstructured data (Kruse et al., 2019), allowing for the analysis of a larger number of variables, thus enhancing the overall quality of the analysis. In short, the capabilities of AI allow financial institutions to provide better services and increase the financial inclusion for more customers, while it can also lead to cost reductions in various areas, such as regulatory compliance.

Meanwhile, AI is deployed in terms of various applications in the banking sector, including smart wallets, voice-assisted banking, banking-process digitisation, and blockchain hosted payments (Donepudi, 2017). Here, smart wallets represent the

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3 advancements in banking services that address the need for customer satisfaction and
4 increased accessibility. Having been widely accepted by various service providers and
5 vendors, smart wallets reduce the dependency on cash payments and decrease the levels
6 of monetary use. Meanwhile, voice-assisted banking largely involves the use of voice
7 commands and touch screens to access banking services (Weill and Woerner, 2015) and
8 accommodates the increasing need for flexibility, especially in choosing suitable regional
9 languages, accessing vital information, and connecting to different financial services.
10 This form of banking is also advantageous since it reduces the possibility of human error
11 while enhancing systemic efficiency. According to Donepudi (2017), digitising the
12 banking processes using AI is also advantageous since it reduces the lines in banking
13 facilities and increases overall productivity. Finally, blockchain hosted payments largely
14 involve capitalising on the capabilities of AI to implement real-time payment processes
15 and speed up the payment procedures, significantly enhancing both customer support and
16 customer satisfaction.

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37 The deployment of AI in the banking sector has largely capitalised on the fact that banks
38 have always been early adopters of technology, with modern technologies having been
39 deployed in the banks for both front-end and back-end applications (Smith and Nobanee,
40 2020). Data is a crucial component in all the business factions of the banks, and the huge
41 data management requirement has accelerated the adoption of AI in banking services.
42 Here, the technology is deployed for autonomous data management services, eliminating
43 the need for human involvement to maximise the speed, accuracy, and efficiency, while
44 it is also deployed for the real-time identification of customers and the prevention of fraud
45 in the realm of online banking (Wei et al., 2012). Indeed, it has presented a crucial
46 initiative aimed at addressing the escalating challenge of credit card fraud, which has
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largely negated the gains made through the advancement of online payments. In terms of being used to verify the identity of clients, the algorithms scan the relevant documents to assess the reliability of the information provided (Smith and Nobanee, 2020). Analysing legal documents and extracting important clauses have also been facilitated via the use of AI technology. Overall, the technology presents a continually advancing system whereby the AI algorithms autonomously analyse data to enhance accuracy.

The introduction of AI in the banking sector has also helped address the systemic challenges associated with the quality of services, especially where making accurate decisions is essential. Here, AI is efficient since it eliminates any possibility of human error (Donepudi, 2017). In addition, the quality of service is mainly assessed in terms of service times, and the use of AI has enabled banks to shorten their service times by overcoming the traditional limitations of the statistic models (Fernandez, 2019). The use of deep learning is another viable intervention that will undoubtedly transform customer services, essentially using specific patterns to predict future activity, predictions that are highly valuable to stockbrokers, asset managers, and investment bankers alike.

The application of artificial intelligence for customer engagement

Chatbots constitute one of the major applications of AI for customer services in the financial services industry, generally being deployed for customer service management and the provision of advisory services (Ba et al., 2010). The use of chatbots in financial services constitutes the aspect of self-service technology adoption, which aims to increase the interaction between the consumer and the technology without necessarily involving the service providers (Jang et al., 2021). Such an intervention is viable in providing customer services for various sectors, including insurance and banking. Chatbots are essentially text-format applications, the popularity of which has been increasing

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3 following the increased use of messaging apps, which have been growing due to the
4 technological advances and the emergence of the millennial generation. In short, the
5 millennials are more attracted to the use of chatbots for customer services since they
6 prefer to chat, which reduces the need for direct conversations (Quah and Chua, 2019).
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14 In fact, this preference for chatting makes it easier to address any customer service queries
15 and other inquiries using AI chatbots (Abdulahi et al., 2021). The chatbots are also
16 attractive due to the indirect nature of the communication alongside other advantages,
17 such as the lack of language barriers, lower operating costs, and 24-hour operations
18 (Klaus and Zaichkowsky, 2020). Indeed, many firms have identified the potential of
19 chatbots in the area of customer services, despite the fact that huge investments are
20 required to develop viable solutions for integration in the legacy system. Financial
21 institutions are the most active in adopting chatbots since they are regarded as a viable
22 tool for facilitating digital transformation. Banks have also introduced chatbots that offer
23 various services to the customers, including checking their account balance information,
24 assisting with their transactions and bill payments, and receiving important
25 communications. Chatbots can also provide vital information to the customers, such as
26 real-time sales notifications, informing users about specific services offered, and making
27 suggestions based on specific contextual information (Belanche et al., 2019). Meanwhile,
28 in the Korean financial industry, chatbots have been deployed for product inquiries,
29 funding purchases and repurchases, making consultations, and conducting overseas
30 money transfers (Jang et al., 2021). In short, the management teams of financial service
31 providers expect that the deployment of chatbots in customer services will improve the
32 overall customer experience.
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3 Virtual assistants can help resolve the customer engagement challenge through
4 conversing with the customers through messaging platforms and mobile applications
5 (Brill et al., 2019), answering customer queries, offering fundamental advice, and
6 addressing common transactions and inquiries (Killa, 2014). Specific AI technology is
7 also deployed in the area of claims management to reduce the overall time taken to
8 process the claims as well as the handling costs while enhancing customer experience.
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10 Indeed, various AI-powered claims assistants have been utilised to speed up the process
11 of verifying and processing insurance claims, with many predicting that this technology
12 will enable instant payments for claims, especially in terms of motor vehicle insurance
13 claims. Certain AI tools are also valuable for risk assessment tasks, often proving vital
14 for speeding up the risk underwriting process (Hall, 2017) through assisting in the rapid
15 analysis of the various demographic components for underwriting, including age, health
16 history, gender, and lifestyle factors.
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35 Meanwhile, AI-based detection systems are suitable for credit card detection systems due
36 to their sophisticated modes of operation and real-time deployment. Automatic credit card
37 fraud detection systems use Bayesian and neural networks grounded in machine learning
38 (Raj and Portia, 2011). Here, artificial neural networks (ANN) are initially trained using
39 the normal behaviour of ordinary cardholders, which allows for the subsequent detection
40 of suspicious activity and the deployment of suitable prevention strategies. Bayesian
41 networks are essentially belief networks consisting of AI programming that combines
42 data mining and machine learning algorithms. Such neural networks are highly
43 advantageous due to their learning capabilities, which eliminates the need for
44 reprogramming. The use of AI in credit card fraud detection is largely based on the high
45 frequency of credit card transactions, which provides the large amounts of data required
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for the training, back-testing, and validation of the algorithms (Bao et al., 2020). The technology also presents a rapid means to combat fraud, avoiding the limitations associated with more prolonged human response procedures. In short, AI is adopted in the field of credit card fraud prevention since it is less costly.

Theoretical Underpinning

The theoretical background of this study is grounded in three fundamental studies. The first was conducted by Mogaji et al. (2021), who devised a conceptual framework pertaining to marketing financial services in relation to vulnerable customers. Overall, the study highlights the ethical implications of data collection and processing, placing huge responsibilities on managers in terms of understanding how their data are being collected and how consumers are targeted using AI and programmatic. The second study was conducted by Ruyter et al. (2018) and revolved around the conceptual exploration of the digital surrealistic approach for deploying AI-based analytics and machine learning to explore customer data and support decision-making. Here, the authors recognised the need to understand the enormous benefits of AI for business operations and how this can translate into actual value for the organisation, which entails the need for managers to understand what AI means and offers as well as how best to use it. The third study involves the development of a conceptual framework by Riikkinen et al. (2018), which presents the prospects of AI in creating value in the field of insurance. Here, the focus was on chatbots and how customer data can be extracted for a value creation that benefits the consumers in terms of product offering and service quality.

Within the context of the present study, the aforementioned theoretical developments are adapted to explore the managers’ awareness of AI and how the technology can be used in their business operations, with a specific focus on exploring the process and any

inherent challenges the managers are facing. Despite the increasing adoption in customer service management, the success of AI in the provision of financial advisory services remains open to discussion (Zhang et al., 2021). The major factors contributing to the lack of adoption relate to the level of expertise, trust, intention of use, and performance expectations (Abdulahi et al., 2021, Soetan et al., 2021). According to Longoni et al. (2019), many consumers hesitate to use AI-based technologies for important and high-risk domains when high-level human expertise is available. This further highlights the need for managers to understand the context of AI, the attendant challenges, and its acceptance by consumers.

The financial service providers have access to the Big Data related to the customers (Dwiledi et al, 2021). There is a growing acceptance among the customers to freely provide their data (Mogaji et al., 2020). However, it is important to recognise some of the major challenges associated with the use of AI, which include consumer privacy issues and technological complexity. Consumer privacy is a major concern due to various legislations pertaining to the security of customer data (Kruse et al., 2019), while the AI applications in the financial services sector entail the major challenge of governance process design (Gillath et al., 2021) and regulatory requirements (Wall, 2018). This notwithstanding, there are huge prospects for AI, and managers must be able to find their way around these challenges. Therefore, the present study was aimed at exploring the lived experiences of managers as they aspire to effectively adopt AI in view of marketing their financial services (see Table 1 for a summary of key studies and contribution of our study).

____ **Table 1** ____

Methodology

Explorative and inductive approach

An exploratory, inductive research approach was adopted for this study to provide insight into managers’ understanding of the use of AI in marketing financial services. This approach was adopted to allow deeper insight into the managers’ work experiences, decision-making processes, and interactions with other stakeholders involved in developing AI systems for their business operations. Specifically, this qualitative approach is relevant for this study since it was aimed at, as Johnson et al. (2006) put it, ‘capturing the actual meanings and interpretations that participants subjectively ascribe to phenomena in order to describe and explain their behaviours’ (p. 132). In fact, this exploratory research approach has been previously adopted in understanding bank managers’ involvement in various business operations. Here, Mogaji et al. (2021) interviewed a number of bank managers in Nigeria to better comprehend their financial inclusion strategies, while Deigh and Farquhar (2021) interviewed various managers to ascertain the impact of their corporate social responsibility initiatives. Larsson and Viitaoja (2017) interviewed Swedish banks managers to explore their banks’ digitalisation processes. According to Bamforth et al. (2018), the qualitative method is suitable for our context as we are in search of an in-depth answer to a specific question, seeking to ‘understand phenomena through accessing the meanings that participants assign to them’ (Orlikowski and Baroudi, 1991, p. 5).

The sample

The target participants are individuals working within the financial services industry in managerial roles, being responsible for sourcing, developing, implementing, and deploying marketing and digital transformation strategies involving AI. The participants were sought across different countries and various financial service providers in terms of

various levels of experience to ensure a diversity of views and experiences to enrich the data and subsequent findings. The managers included not only individuals working in banks but also individuals working in Fintech and payment services companies. Great efforts were made to reach out to different managers across different countries to ensure a multi-country approach was applied in the study. Here, we targeted participants from the UK (Europe) and Canada (North America) to represent developed countries and from Nigeria (Africa) and Vietnam (Asia) to represent developing countries and emerging markets. The participants were initially contacted through the personal networks of the research team with the use of snowballing sampling. In addition, potential participants were targeted through LinkedIn, specifically those with managerial roles in finance, marketing, and information technology across the four different countries. One hundred twenty-seven prospective participants were contacted through the inMail function on LinkedIn. They were then provided with details regarding the research context and the background of the study before their participation was requested for interviews that would be carried out online. We recognise the possibilities of self-selection bias in our sample and made an effort to address this by seeking a defined target population and a sampling frame, contacting different managers from different countries, and following up on non-responders, making the study representative by including as many people as possible (Young et al., 2020, Czarnecka & Danbury, 2018). In total, 31 participants responded and confirmed their willingness to participate in our research project. An additional 16 participants were recruited through personal contact, making the total number of participants 47, all of whom were managers working in the four countries. The participants' demographic information is presented in Table 2.

_____ **Table 2** _____

Data collection

Qualitative data were collected through semi-structured interviews conducted with the participants. The interview questions were developed based on the existing literature, with a specific focus on the conceptual frameworks developed by Mogaji et al. (2020), Riikkinen et al. (2018), and Jang et al. (2021). The questions were presented in an open-ended form to allow the participants to express themselves and increase the possibilities of capturing more emerging ideas (Bamforth et al., 2018). The questions were tested using a pilot study with six managers who were not part of the final sample. The pilot study allowed us to refine the scope of the questions, improve the manner in which the questions were asked, achieve clarity and monitor the timing, duration, and flow of the interviews. The pilot interview participants did not identify any serious flaws or any sensitive issues with the interview protocol but did provide some feedback on the type of questions and the flow of the interview. In response to their feedback, we made a few changes to the wording, format, and arrangement of the questions.

Given that the participants were in different countries and time zones, the interviews were conducted virtually using Zoom. This part of the work was completed by the first author. The interviews began with a brief explanation of the objective of the study and the ethical considerations. Even though the participants had signed a consent form via email, verbal consent for their participation and for recording the interview were sought prior to the interview on Zoom. The participants were informed that they could leave the interview at any point without giving any reasons, and they all agreed that the interviews could be recorded to aid the transcription and allow us to better understand their comments.

The participants were asked the same broad and open-ended questions about their understanding of the use of AI in financial services, their roles in developing the AI

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3 systems in their organisations, their efforts in deploying AI for marketing financial
4 services, and the challenges they have faced in integrating AI in these services. The
5 interview session also provided the managers with an opportunity to ask probing,
6 clarifying questions to expand on any emerging themes, while they were also allowed to
7 ask questions at the end to allow for a more comprehensive discussion (Bamforth et al.,
8 2018; Farinloye et al., 2019). The interviews lasted between 58 and 83 min (median 67
9 min), with the recorded interviews transcribed by a professional and saved in PDF form
10 for further analysis. Additional notes were taken during the interview and were included
11 as part of the dataset for further analysis.
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23 24 25 *Data analysis*

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27 The transcribed data were analysed following the thematic analysis guidelines suggested
28 by Braun and Clarke (2006). This began by initially reading the transcribed interviews
29 intensively and repeatedly to gain a better understanding of the participants' responses
30 (Farinloye et al., 2019). This was followed by uploading the interview transcripts into
31 NVivo, a qualitative data analysis (QDA) computer software package. The transcripts
32 were reread using the software, and the data were carefully examined to identify any
33 related concepts and categories (Deigh and Farquhar, 2021). Meanwhile, Auto-transcribe,
34 an automated transcription technology function used to assist in the analysis of extensive
35 datasets, was also run on our dataset. Following the auto-transcription, the transcripts
36 were subsequently manually annotated using the software. The editor function was used
37 to review and change the transcripts, highlight relevant codes and create specific nodes
38 as they emerged from the dataset. These codes were subsequently grouped into relevant
39 themes that pertained to the objectives of the study. The frequency of the themes across
40 all the transcripts indicated how important they were. A number of the themes were also
41 expanded to reflect new emerging themes, while others were removed or merged since
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they did not have enough content and quotes to stand alone as core themes. At this stage, any hierarchical and associative relationships among the codes and themes were also established (Soetan et al., 2021). The final themes and sub-themes were identified and shared with other research teams and professional colleagues to provide a quality check and additional insights. No further changes were made at this stage, and the 23 sub-themes were merged into five key themes pertaining to the three main objectives of the study.

Data credibility

Considerable efforts were taken to ensure the credibility of the data. First, the ethical guidelines of the authors’ affiliated institution were followed, including in terms of consent, data security, and confidentiality (Deigh and Farquhar, 2021). Second, we verified the role of the participants to ensure that they were giving us the right information based on their roles. This information was verified using LinkedIn, any news coverage announcing their appointment or media coverage of their business operations, as well as the participants’ profiles on their company websites. Third, the participants were allowed to control their involvement in the projects, which ensured that they were willing to give information that they were comfortable with and not forced or put under undue pressure to disclose any information. Fourth, we shared the transcript interviews with the participants to ensure that we had correctly covered and transcribed the information they had provided (Czarnecka & Mogaji, 2020). This is described as a ‘member check’ and is regarded as one of the main ways to ensure the credibility of a qualitative study (Merriam and Tisdell, 2015). Fifth, the different stages of the data analysis were presented through an audit trail and a rigorous peer debriefing process through managing the process with the team and receiving quality assurance of the data from other colleagues (Miles et al., 2013). Sixth, based on the in-depth description recommended by Lincoln and Guba

(1985), we provided a detailed description of the quotes from the interviews to reinforce each point.

Findings

The qualitative data analysis revealed three key themes surrounding the managers' understanding of the use of AI in marketing financial services. While there were variations in terms of the experience based on the type of financial services, the years of experience, and the participants' country, the main themes were as follows: (1) awareness of the need for AI, (2) addressing the growing need for AI in their business operations, and (3) accelerating the adoption for enhancing business operations. These themes are discussed below and supported with quotes from the managers.

Awareness regarding artificial intelligence

The managers recognised the importance of AI as a tool for business transformation and how it would change their business operations, highlighting that AI is becoming an integral part of the conversation surrounding digital transformation and consumer engagement within the financial services industry. The participants also noted that AI is not only about marketing but also about developing financial products and services, such as cryptocurrency, blockchain, and financial engineering. For example, one of the managers working with a commercial bank in Canada alluded to the idea that 'AI is now vast in financial services, involving the design and marketing of financial services'.

However, the managers also raised a concern regarding the ongoing misconceptions about the prospects and possibilities of AI, with many people remaining unsure about AI and continuing to confuse the basic technology used in banking with AI, noting that everything could not just be AI. One manager described it as follows:

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3 *[AI is] a buzzword that everyone is using and talking about, but few people know*
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5 *what it is and how it actually works. Everyone has a different understanding of*
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7 *AI, depending on their level of experience and exposure.*
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10 These misconceptions were also highlighted among the managers from the developing
11 countries, who noted that not many people are aware and involved with AI and can often
12 simply claim they know how it works. Some of the managers even noted that not everyone
13 who works within financial services knows about AI and how it truly works.
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21 Chatbots also emerged as one of the most common examples of AI used in financial
22 services. Here, the managers recognised that having a chatbot on WhatsApp or Messenger
23 to engage with the customers is like an entry-level form of AI in financial services, while
24 they did acknowledge that this type of application had transformed financial service
25 provisions. However, they noted that there is more to AI than merely chatbots, noting that
26 a chatbot is just one of the many possibilities of AI. One participant from the UK stated
27 the following:
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37 *[h]aving a chatbot on your website is now very easy and it is becoming a*
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39 *standard; but you need to know that chatbot is not all, that's just about*
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41 *communicating; ultimately, you need to be able to collect data from the chatbot*
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43 *and use it to inform your business strategy.*
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47 This was further corroborated by another participant from Nigeria:

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49 *...chatbot is a good way to start but we know that is just the entry-level. We are*
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51 *still doing more with AI in the banking sector.*
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56 Overall, the managers acknowledged that chatbots appear to be the only physical aspect
57 that the customers are aware of when it comes to the AI used in financial services, and
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there is little awareness on the part of the customers of the various other algorithms working behind the scenes and influencing the decisions for the customers.

The awareness that there are many more possibilities for the use of AI in financial services ensured that the managers recognised the importance of exploring other options to meet customer needs and improve their knowledge of AI. In short, the managers realised that they need to increase their awareness about AI, what it offers, and how to adopt it for their business:

[w]ith the many misconceptions about AI and the growing drive for digital transformation in this sector, it is important for staff to be very conversant with this technology, or at least to know the theory and how it really works.

Addressing artificial intelligence as a business need

Following on from the awareness of the prospects of AI, the managers shared their thoughts regarding the measures being put in place to integrate AI in view of addressing their business needs. Here, one of the managers noted the following:

...it's not just about recognising how it works but, importantly, seeing how it works in your company. You want to try things out and improve the customers' experience.

While many managers agreed that the need for AI in their operations appears to evolve automatically and naturally as the need arises, some could not identify the exact time they decided to use AI. However, the integration of AI in business practices evolved as part of their organisations' transformation strategies. Three managers working in Nigeria noted that they had observed the growing trends, had shared them with their managers and had provided some direction on how to go about addressing them, while seven of the participants noted that the idea for AI was pitched to them by various Fintech companies.

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5 Ultimately, the managers identified three main challenges in addressing the business
6 needs for AI: convincing the management team that it is needed, assembling the right
7 team to do it, and deploying the system and making sure it works within the existing
8 infrastructure. The managers also noted a need to justify that the AI system is necessary
9 for effective marketing and delivering financial services, which often includes sharing
10 examples of other banks – both within the country and outside it – that have adopted AI.
11 The managers further reiterated that the lack of awareness about AI at the management
12 level also presents a challenge for the quick adoption and deployment of AI applications,
13 with one manager from Nigeria stating the following:
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26 *...when your boss does not know much about these things, it becomes harder to*
27 *convince them that it will work; sometimes, they think it's an idea of the*
28 *millennials and therefore consider it to be expensive and extravagant.*
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35 Even when they have received authorisation from the banks' management team to
36 continue developing the AI system, the managers felt that assembling the team to work
37 on the project could be challenging, especially for the marketing managers who often feel
38 that they lack the technical abilities to get involved in the project or even lead the team.
39 Here, the managers noted that they often invite the Fintech companies that originally
40 pitched the idea to collaborate with their in-house team at the bank, while, in other cases,
41 they sought out another team that could do the job better and possibly is more cost-
42 effective and scalable, or hired additional staff members to aid the in-house team in
43 completing tasks.
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Irrespective of the approach taken to assemble the team, the managers also noted that managing a team comprised of individuals with varying technical capabilities could pose a challenge, especially when the marketing team shares their expectations and the developers state what is realistically possible. The managers shared various instances of when collaboration between the teams became difficult and how they brought in a consultant to help resolve the issue. The managers further reiterated the conflict between their marketing expertise and collaborating with tech developers, albeit that, ultimately, they all want the same thing.

Accelerating the adoption of artificial intelligence

The managers acknowledged that AI in marketing financial services is here to stay and that it is, therefore, vital for them to accelerate its adoption to enhance their business operations and maximise profits through enhanced services. They did, however, recognise the role of regulators in achieving this, emphasising the importance of the regulators in helping to ensure that all stakeholders are kept informed. Here, the participants in the UK acknowledged the good works of the Financial Conduct Authority (FCA), the regulatory body of UK banks, and what this conduct regulator was doing to raise awareness about the implications of AI in financial services. One participant noted the following:

[t]he FCA is not just interested in the marketing but the overall impact of AI in financial services, including in terms of data quality and ethics.

However, this is not the case for many of the managers from the developing countries, who had varying perspectives and expectations of their regulators, with one participant from Nigeria stating the following:

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3 *I don't think the CBN [Central Bank of Nigeria] is aware of AI financial services*
4 *as I have not seen any circular on information from them; the last thing I heard*
5 *was about the cryptocurrency ban, but we know that's not really about AI.*
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12 The managers also highlighted the need to ensure the customers 'buy-in' to accelerate the
13 use of AI, recognising the value of data and the importance of encouraging consumers to
14 provide additional data and, importantly, to adopt the technological innovations that are
15 being offered. The managers observed an increasing level of advertisement and marketing
16 communications aimed at informing people about chatbots and other forms of digital
17 transformation. In addition, the managers believed that the customers should have
18 confidence that banks are looking out for the customers' best interests.
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30 The bank managers themselves also stated that they need to acquire more knowledge and
31 understanding of AI and how it affects their e-business and operations as it becomes
32 increasingly necessary for them to engage in conversation. The managers also suggested
33 that this knowledge acquisition regarding AI should also extend to the top management
34 level as this could assist in driving down costs and cascading the digital transformation
35 strategies.
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47 The human resource aspect is also instrumental in accelerating the adoption of AI in terms
48 of bringing onboard individuals who understand what is appropriate, working with the
49 right technical partners, and ensuring that matters are conducted ethically. One participant
50 from Nigeria noted that,
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55 *...the last project we did had to be outsourced to some guys in India; we had little*
56 *control over the design process; we simply trusted that they did a good job.*
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3 These participants acknowledged that having individuals with the right skill sets to do
4 this job is important. Here, another participant noted the following:

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7 *[w]e must acknowledge that we are marketers, we are not developers. We always*
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10 *need to have the right team to get things done.*

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12 This recognition aligns with earlier comments related to improving the team, which
13 includes outsourcing the project to a competent team or recruiting more individuals to
14 bolster the capabilities of the in-house team.
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21 The country in which the organisation operates also influences the extent to which AI
22 adoption is accelerated, with the data analysis revealing a significant difference in
23 experience between the participants from developed countries and those from developing
24 countries. Specifically, the participants from the developed countries were more
25 confident about the trajectory of AI, with one participant from the UK stating the
26 following:
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35 *[v]ery soon, we won't need high street banks; all banking operations will be*
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37 *carried out on mobile phones.*
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40 In contrast, the participants from the developing countries were more pessimistic, being
41 largely concerned about the regulatory framework, the level of technological
42 development in their country, the consumers' attitude towards technology, the data and
43 infrastructure required to support AI algorithms, the size of their financial services
44 industry, and the manpower required to champion the innovative ideas. One participant
45 from Nigeria concluded with the following:
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54 *[w]e know the world is moving fast with AI, but we will catch up at our own pace.*
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Discussion

According to our qualitative study, the bank managers – both from the developing and the developed countries – were aware of the prospects of AI in the area of marketing financial services. They all recognised that AI had brought a much-needed digital transformation into the provision of financial services. While the role of chatbots is well recognised in this provision, the managers recognised that they are not the only type of AI system needed and highlighted a variety of other ways to integrate AI into financial services marketing. There remain, however, various concerns related to managers' expectations regarding the development of AI algorithms, the role of the regulators, and the inherent challenges pertaining to the consumers' adoption of AI systems.

Theoretical contributions

While we recognise the variety of work on how AI is shaping marketing practices (e.g., van Esch and Stewart, 2021; Davenport et al., 2020; Dwiledi et al., 2020, Wayne et al, 2020), this study specifically contributes to the areas of marketing that are focused on AI and marketing financial services. Specifically, we provide insight that extends the work of Riikinen et al. (2018) on using AI in creating value in insurance service provision as well as that of Mogaji et al. (2021) on theoretically positioning the implications of AI in terms of digital marketing of financial services to vulnerable customers. Beyond the conceptualisation and synthesis of the existing academic literature, this study provides empirical insight into the opportunities, prospects, and challenges pertaining to the adoption of AI in the field of financial services marketing. (see Table 1 for a summary of key studies and contribution of our study).

Our study further presents a conceptual framework of AI in relation to financial services marketing. The framework presented in Figure 1 identifies various key constructs and

stakeholders related to adopting AI for financial services marketing. This framework and its implications build on the analysis of the qualitative data collected during the interviews with managers as well as the existing theoretical studies, such as those from Mogaji et al. (2021), Riikkinen et al. (2018), Dwivedi et al. (2021) and Ruyter et al. (2018), as identified in the theoretical underpinning section of this study.

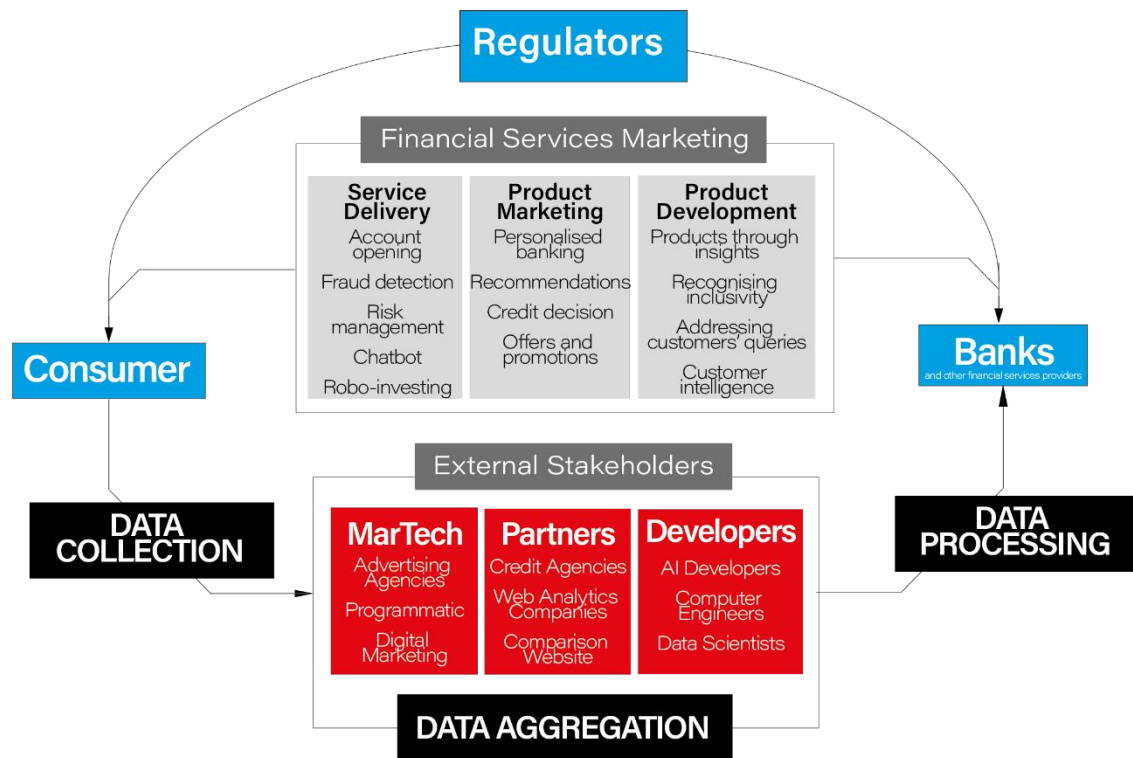


Figure 1. Conceptual framework of the use of AI in financial services marketing

The main constructs here are data collection, aggregation, and processing. Our framework demonstrates that data are being collected from the customers to develop AI algorithms for banks and other financial services providers. The data are either willingly provided by the consumers when opening a bank account or searching for products on comparison websites or through leaving a digital footprint via credit agencies and government organisations (Mogaji et al., 2020; Riikkinen et al., 2018). Key external stakeholders are presented here as being responsible for aggregating the collected data, which may not

necessarily be the bank managers. For example, marketing technology (MarTech) companies may integrate certain elements of advertising agencies and engage with digital marketing and programmatic to deliver advertisements to individuals based on their browsing history or existing information (Mogaji et al., 2019). In addition, there exist various partners responsible for aggregating this information and making it useful for the banks and other financial institutions. For example, credit agencies such as Experian collect and aggregate information from millions of people in the USA, the UK, Brazil, Colombia, and India, and assist organisations worldwide in lending in a more responsible, fairer, and rapid manner to both individuals and businesses (Experian, 2021). Moreover, web analytics companies such as Google (Analytics), Facebook (Insights), and Crazy Egg, as well as the various comparison websites, collect data from customers that are subsequently fed into the data aggregation process. These aggregate data are made available for developers – including data scientists – to develop algorithms that can shape the marketing and availability of a given financial product. In addition, it is crucial to recognise the need for data processing to ensure that the data collected by the banks and external stakeholders are presented in a format that can be used to make business decisions.

The information owned by banks (voluntarily provided by customers) and the aggregated data collected from external stakeholders are being used for service delivery purposes, which includes introducing customers to a different account, detecting fraud when a new account is opened, and providing relevant answers to customers’ queries via chatbots. Here, we observed that bank managers may lack control due to the inherent design and development of the algorithms, which are frequently performed by the developers and partners. Accordingly, there are certain prospects for marketing financial service

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3 products, especially in terms of personalised banking, recommendations, offers, and
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5 promotions. These are often presented automatically as a result of machine learning and
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7 AI's understanding of customers' past financial transactions and their current financial
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9 state. Lastly, as part of the financial services marketing opportunities, this information is
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11 being used product development and ensuring the right customers are introduced to the
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13 right product. This enables banks to have an overview of their business operations and to
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15 identify how they can best serve their customers.
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22 As reported by our participating bank managers, the role of the regulators cannot be
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24 overlooked, especially as they serve as control measures between the engagement of the
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26 consumers and the banks. The managers acknowledged that the regulators are in place to
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28 provide overarching support and oversee the marketing process, as well as to ensure that
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30 the industry benefits from the innovation brought about by AI. These responsibilities of
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32 the regulators are important for two reasons. First, as always having been part of a
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34 regulated industry, even prior to the advent of AI and the digital transformation, financial
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36 services providers are expected to regulate their marketing strategies (Mogaji, 2018) to
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38 ensure that all the adverts are 'clear, fair and not misleading, regardless of the media type'
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40 (FCA, 2021). Second, AI has brought additional concerns regarding the monitoring of the
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42 practices of financial services providers. Here, the FCA acknowledges that the use of data
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44 analytics and AI is growing and must be monitored to protect the customers' interests
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46 (FCA, 2021b). Meanwhile, Canada's Office of the Superintendent of Financial
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48 Institutions (OSFI) also launched a consultation on technology risks in the financial
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50 sector, highlighting the need for a regulatory and supervisory framework (OSFI, 2020).
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52 This highlights the growing interest among regulators in monitoring the impact of digital
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54 technology on service provision. However, the managers in the developing countries
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expressed certain reservations about the role of the regulators and how this presents certain practical implications, both for the industry and the consumers.

While this conceptual framework illustrates the key stakeholders, perhaps more importantly, it challenges three existing notions about AI and financial services marketing.

First, there is a distinction between AI and financial services and the use of AI in marketing financial services. Here, AI can be applied in developing financial services and products, but deeper insight can be used in developing marketing strategies for financial services (See Appendix 1). The application of AI in financial services entails specific computer knowledge and an awareness of how AI can be used for developing financial services. However, this is often not what financial service marketing managers are expected to do. In fact, AI and financial services involve the use of Fintech, which is often focused on blockchain and cryptocurrencies, and while these are essentially part of financial services, they are decidedly distinct from the marketing aspect of those services. In short, they place a premium on ‘marketing’ the ‘financial products’ using huge computing power, access to Big Data, and the analytical capacity of AI to raise awareness about financial products and to make them available and accessible for all prospective customers.

Second, the adoption of AI in financial services encompasses more than simply chatbots. Often, the focus of the AI in the area of financial services marketing has been on providing chatbots and facilitating customers’ engagement with them (Abduladri et al., 2021; Mogaji et al., 2021; Riikkinen et al., 2018). However, while this is undoubtedly important

for service delivery, it is crucial to recognise that a chatbot is only one of the numerous digital tools that allow customers to interact with the bank and obtain information. Beyond chatbots, the AI in financial services marketing entails analysing customer behaviour, marketing the financial services to individuals who need them, and developing better ways to provide prompt and rapid decisions on credit applications. Therefore, practitioners and marketers must take a holistic approach to the AI used in marketing and explore other options to enhance their business operations.

Third, our study reveals that the role of bank managers in developing AI for financial services marketing remains somewhat limited. Following our interactions with the managers and the insights gained from the conceptual framework, it is clearly important to separate the job description of the banks' marketing managers and that of the Fintech developers. Often, the managers are involved in developing these systems by sharing their business expectations and allowing the developers to develop ideas that can help them actualise them. Meanwhile, the developers devise solutions that can be applied to different business sectors. For example, a chatbot developer can develop an application for a bank *and* a university. It is, however, essential to recognise the working relationship between the bank managers and the developers. Without doubt, developers require the involvement of the marketing managers to improve their products and their ability to meet the business needs of banks.

Managerial implications

This study provides practical implications for managers, policymakers, and other stakeholders. Table 3 presents a summary of managerial implications. First, banks need to understand their business objectives, resources, and customers' needs (de Ruyter et al., 2018; Gökerik et al., 2018). It is important to recognise that AI has a wide variety of

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3 applications and that one size does not fit all. Much like its customers, each bank is
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5 unique. The fact that every company is adopting AI and that it is becoming a business
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7 buzzword does not mean that a company should rush into using it. It is critical for banks
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9 to understand the different options and how they fit their specific business needs.
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14 Second, after determining whether AI is required and for what purpose, managers should
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16 consider the full range of development options available to them to meet their specific
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18 business needs. Managers should reflect on the importance of assembling the right team
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20 to develop the product, should be mindful of the key performance indicators that will be
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22 set to gauge the success of the projects, and must consider the prospects of managing and
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24 maintaining the algorithms integrated into their business practices. While technology
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26 development may know no geographical background, managers should consider the pros
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28 and cons of collaborating with the developers in their country, especially the developing
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30 countries, or working with a global team.
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37 Third, in selecting a team or working with partners, managers should be mindful of the
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39 ethical implications of their working relationship. This is important in terms of the
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41 collection, aggregation, and processing of data. As Mogaji et al. (2020) highlighted, when
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43 dealing with digital marketing and data harvesting from vulnerable customers, managers
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45 should be mindful of the short- and long-term implications of how the data are collected
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47 and processed for machine learning and the subsequent development of AI algorithms. In
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49 short, the data should be fairly and ethically collected, aggregated, and processed.
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56 Fourth, bank marketing managers should be trained and supported in understanding the
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58 AI used for financial services. Trained and educated managers will also have a better
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working relationship with their teams and partners, as well as be more conversant with the technology, allowing them to influence the design process. In addition, these managers will be able to challenge any bad practices and ensure compliance with the regulatory requirements.

Fifth, the relevant regulators must be involved in the development of AI for financial service marketing. In short, it is vital to have a regulatory framework in place to guide the design process and ensure that customers are treated fairly. Furthermore, managers should implement measures to mitigate the risks posed by their AI-based systems (Gow, 2021). As a number of our participants in the developing countries expressed certain concerns about the regulatory requirements and awareness of AI in their country, they must consider regulatory advice from other governmental bodies, such as the European Union's ethics guidelines for trustworthy AI (EU, 2019), and from tech developers such as Microsoft, who have provided key AI-related ethical principles pertaining to fairness, inclusiveness, reliability and safety, transparency, privacy and security, and accountability. In addition, AI systems should involve algorithmic accountability to enable appropriate human direction and control (Microsoft, 2017).

Lastly, communicating the prospects of AI to consumers is also crucial. In short, consumers should be aware of the technology and how it can be used to enhance their banking experience. Here, Mogaji et al. (2021) and Abdulhadi et al. (2021) found that the banks in developing countries were not doing enough to communicate the availability of chatbots to the consumers, which has influenced their adoption. Consumers should be made aware of the benefits of using chatbots and be reassured that the AI system is secure in supporting their transactions. In addition, according to the existing ethics guidelines,

consumers should be assured of appropriate data privacy and governance (EU, 2019), should be informed about how their data are being collected and aggregated to develop financial products and should be reassured that there exist data governance mechanisms to support the data collection, aggregation and processing.

_____ **Table 3** _____

Conclusion

This study was aimed at exploring managers’ perceptions of the adoption of AI in the area of marketing financial services. With the growing prospects of AI and the digital transformation of business practices in the financial services industry, managers must be aware of the prospects, opportunities, and challenges associated with the AI used in the industry. The study makes a significant theoretical contribution to the existing works on AI and the financial services market by introducing a theoretical framework that identifies the key constructs and stakeholders in integrating AI for financial services marketing. Specifically, the study challenged certain existing notions about AI and financial services, the chatbots used for financial service delivery, and the role of marketing managers in developing AI.

As with any study, the present study is not without its limitations, and the findings should be interpreted in light of these. A qualitative-based methodology was adopted for the study, which involved the self-reporting of the managers, meaning the findings may not be generalisable. Individuals with different roles and levels of knowledge of AI who reside in different countries may have other experiences of adopting AI in their business operations. Future studies can endeavour to adopt a quantitative approach to fully quantify the level of bank managers’ understanding. Finally, the theoretical framework

developed is open to further research and validation in view of establishing whether the constructs are valid and applicable to different business operations.

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Table 1 - Summary of key studies and contribution of our study

Theme	Key Study	Aim	Methodology	Key Findings	Contribution of our study
Marketing and AI	Dwivedi (2020)	Multi-stakeholder perspectives on the significant opportunities and prospects of AI in marketing and other business practices	Conceptual paper/ Viewpoints	Growing access to bid data and computing capabilities which will shape the future of the industry and society at large	Most of these studies were conceptual and viewpoint. Our present study provides insight into the practitioners' perspective of Ai adoption in marking financial services.
	Davenport et al. (2020)	Postulated how AI would change the future of marketing, citing the algorithm's ability to automate business processes, analyses big data, and gain insight from customers and transaction data.	Conceptual paper	Proposed a multidimensional framework for understanding the impact of AI in marketing	Beyond conceptualising and synthesising academic literature, the study provides empirical insight into AI's opportunities, prospects, and challenges in financial services marketing. We provided insight into different industries and different countries to understand the drivers and inhibitors of a successful AI adoption in marketing financial services.
	Mogaji et al. (2020)	Proposed the possibilities of using AI to personalise emotionally appealing advertisement	Conceptual paper	AI can be developed to collect data from customers (including their digital footprints) to create a personalised advertisement	Beyond the conceptual approach of these papers, we provide qualitative insight from the marketing managers.
	Mogaji et al. (2021a)	The implications of artificial intelligence on the digital marketing of financial services to vulnerable customers	Conceptual paper	Ethical collection of data is essential. Effectively targeting the customers is also necessary. There should be a form of human control and empathy in the decision-making process.	

	Ma, L., & Sun, B. (2020).	Provides conceptual insight for marketing researchers and practitioners to leverage machine learning methods for business operations.	Conceptual Paper	Identified several marketing research priorities, highlighting opportunities for machine learning methods to be integrated into marketing research.	
Chatbots	Jang et al. (2021)	Explore the managers' perceptions of chatbots in South Korea	Semi-structured interviews with bank managers	Managers perceive the chatbot services to meet customer expectations.	Our study extends the works of Jang et al. (2021) to explore managers' perceptions of chatbots (and AI at large) across different countries and financial services providers. Our study recognises that AI in financial services encompasses more than chatbots. While chatbot is essential for service delivery, it is crucial to remember that it is just one of the numerous digital tools that allow customers to interact with the bank and obtain information. While previous studies have often focused on a particular country, our study has a holistic view of digital transformation across the industry and different countries. Our study revealed that though bank managers are interested in having chatbots for enhancing financial services provision, they lack control due to the developers' inherent design and
	Quah & Chua (2019)	Investigates the effectiveness of the current use of chatbots in Singapore.	Literature review and interviews	Chatbots increase the operation of banking services and reduce operational costs.	
	Riikkinen et al, 2018	Analysed how insurance chatbots support customers value creation	Literature Review	Chatbots constitute a new interaction medium that can be used to influence value creation.	
	Eren (2021)	Investigates effects of customers perceived trust in bank's chatbots. The focus was on Turkish consumers.	A quantitative study, a survey of bank customers	Perceived performance, perceived trust and corporate reputation significantly affect customer satisfaction with chatbot use.	
	Abdulquadri et al. (2021)	Explores how Nigeria banks are developing chatbots as a digital transformation tool to radically change business models,	The Search-Access-Test (S-A-T) model was adopted to understand	Banks are adopting chatbots, but it is essential to consider reassuring consumers about the chatbot's security and capabilities.	

		improve customer experience and enhance financial inclusion in emerging markets.	how Nigerian banks are adopting chatbots.		development of the algorithms.
	Mogaji et al. (2021b)	Explores how consumers in an emerging market interact with banking chatbots	Qualitative study. Semi-structured interview with bank customers	Consumers are interested in chatbots, but they want a responsive chatbot with a beautiful interface and secured features.	
AI in Financial Services	Zhang et al., (2020).	Investigates differences in consumers' perceptions of trust, performance between humans, financial advisors and robo-advisors. The focus was on consumers in the United States	Experiments	Consumers prefer human, financial services to robo-advisors.	While studies are often conceptual, based on secondary data and literature review, while still focusing on one country, our study provides empirical insight through qualitative data collected from managers and financial service providers across different countries.
	Klaus & Zaichkowsky (2020)	Documenting how AI has transformed customer decision making.	Conceptual paper/ Literature Review	Consumers focus on convenience, time saving and effectiveness.	Our study presents a conceptual framework of AI in financial services marketing. The framework identifies key constructs and stakeholders in adopting AI for financial services marketing.
	Raj et al. (2011)	Analysis of techniques used in credit card fraud detection.	Conceptual paper/ Literature Review	Modern credit card fraud detection technologies are based on AI, machine learning, genetic programming and sequence alignment.	Our study presents a distinction between AI and financial services and AI in marketing financial services. AI can be applied in developing financial services and products, but also, a more profound

	Bao et al. (2020)	Review challenges in detecting fraud using machine learning.	Conceptual paper/ Literature Review	Machine learning is a promising intervention for fraud detection. Huge prospect for continued use in financial services provision.	insight can be used in developing marketing strategies for financial services.
	Kelley et al. (2018).	Analyses the impact of AI on the workforce in the insurance industry in the United States.	Conceptual paper/ Literature Review	AI will be a game-changer for industries to transform how the insurance industry works.	We found that the role of bank managers in developing AI for financial services marketing is still limited. Our study posits that it is essential to separate the job descriptions of banks' marketing managers and Fintech developers
	Kruse et al. (2020)	Analyses the challenges for AI in the financial services industry in Germany.	Conceptual paper/ Literature Review	The adoption of AI faces numerous challenges associated with critical success factors.	Given that financial services have always been a regulated industry and with AI bringing additional concerns about monitoring the practices of the financial services providers. Our study reiterates the role of the regulators, primarily as they serve as control measures between the engagement of the consumers and the banks. There are also implications for regulators in developing countries.
	Wall (2018)	Analyses the financial regulatory implications of AI in the United States.	Conceptual paper/ Literature Review	The rapid development of AI will transform the financial services sector.	

Table 2. Demographic information of informants

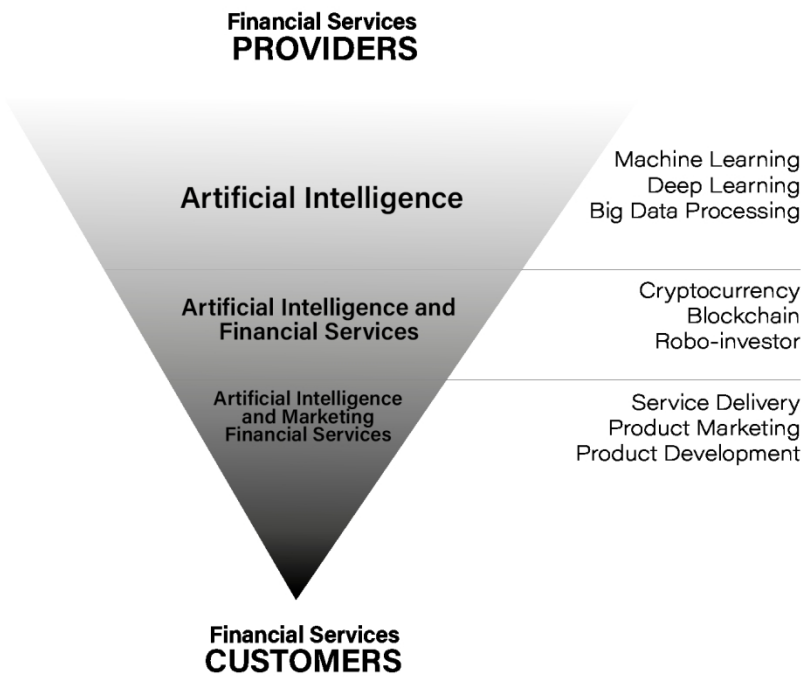
Characteristic	Subgroup	Frequency	%
Gender	Male	27	57.4%
	Female	19	40.4%
	Non-binary	1	2.1%
Age	20- 29	9	19.1%
	30-39	17	36.2%
	40-49	14	29.8%
	50-59	5	10.6%
	60+	2	4.3%
Country	Canada	10	21.3%
	Nigeria	11	23.4%
	United Kingdom	14	29.8%
	Vietnam	12	25.5%
Industry	Non-retail bank	19	40.4%
	Islamic bank	2	4.3%
	Fintech/NeoBank	12	25.5%
	Credit card	5	10.6%
	Insurance	9	19.1%
Years of experiences	0-9	15	31.9%
	10-19	25	53.2%
	20 +	7	14.9%

Table 3: Summary of Findings and Managerial Implications

Key themes and Implications	Sub Themes	Sample Quotes
Awareness about AI <i>Managerial Implications</i> - Bank marketing managers should be trained and supported in understanding the AI used for financial services. - Trained and educated managers will also have a better working relationship with their teams and partners - Managers should communicate the prospects of AI to consumers. - Consumers should be aware of the technology and how it can be used to enhance their banking experience. - Consumers should be made aware of the benefits of using chatbots and be reassured that the AI system is secure in supporting their transactions. - Consumers should be assured of appropriate data privacy and governance - Consumers should be informed about how their data are being collected and aggregated to develop financial products - Consumers should be reassured that there exist data governance mechanisms to support the data collection, aggregation, and processing.	The Hype	<i>[AI is] a buzzword that everyone is using and talking about, but few people know what it is and how it actually works. Everyone has a different understanding of AI depending on their level of experience and exposure.</i> <i>It is very broad and involves many technicalities that can be very confusing to people and that's why people keep confusing AI with machine learning and deep learning. There are a lot of misconceptions out there.</i> <i>...just like any technology, people struggle to understand it and we all learn on the job; likewise with this AI, many people don't know about it yet.</i> <i>[w]e all hear about it, we read about it, and we see how companies are using it. We all know this is important but not very many people know much about it and how to go about it. Often, you will need the tech guys to sort something out for you.</i> <i>[c]onsumers don't know how many algorithms are involved in deciding if they will get a loan or even the interest they will be offered; they do not know much about the influence of these systems in deciding which type of product they are offered; a chatbot is just a part of the system and there are many other hidden values of AI not recognised by the customers - UK retail bank.</i> <i>Many customers are not aware of the AI used in financial services and not to mention chatbot. There</i>
	The misconceptions	
	The Chatbot.	
	The huge possibilities	

		<i>are many things going on with technology in the banking sector – Nigerian Bank Manager</i>
Addressing the Business Need <i>Managerial Implications</i> • Banks need to understand their business objectives, their resources and their customers’ needs • Managers should reflect on the importance of assembling the right team to develop the product • Managers should consider the pros and cons of collaborating with the developers in their country, especially the developing countries, or working with a global team. • Managers should be mindful of the ethical implications of their working relationship with developers and other suppliers. • This is important in terms of the collection, aggregation, and processing of data. • Managers should be mindful of the short- and long-term implications of how the data are collected and processed for machine learning and the subsequent development of AI algorithms.	Addressing AI as a Business Need	<i>[h]ighlighting how our competitor is using AI often works in terms of convincing the management team that we should do the same. You must recognise that no bank wants to lose out due to this digital transformation – Vietnamese Bank Manager</i> <i>[o]ne of these Fintech companies pitched an idea for us. We thought it was nice and the team liked it as it would complement what we already had, but the pricing structure was not making much financial sense; we had to see if we could do the same in-house with our team, but we found that it was not feasible, so we finally agreed to get another tech company to work with us – UK Manager</i> <i>I must acknowledge my weakness at this point with regard to this technology. I was on the team to make sure we integrated the AI system for credit scoring and decision-making, but it was not a good experience. We had the idea of what we wanted to do, but the team couldn’t just come up with something worthwhile. I tried to engage with some of the terminologies and technicalities, but it seems we were on a different threshold – Nigerian Bank Manager.</i>
	Evolving naturally	
	Convincing the management team	
	Assembling the right	
	Deploying the systems	
	Managing the team	
Accelerating AI adoption <i>Managerial Implications</i> • Managers should consider the full range of development options	The Regulator’s role	<i>[c]onsumers need to be assured that we are working to develop a system that works for them and for the banks. Consumers should trust the bank in that the decisions being made through the system are right and unbiased – Canadian Manager</i>
	The consumers’ trust	
	The managers knowledge	
	The manpower	
	The country of operations	

available to them to meet their specific business needs. • Managers should ensure that data are fairly and ethically collected, aggregated and processed. • Trained and educated managers will be more conversant with the technology, • Managers will be able to challenge any bad practices and ensure compliance with the regulatory requirements. • Relevant regulators must be involved in the development of AI for financial service marketing. Regulatory framework should be in place to guide the design process and ensure that customers are treated fairly. • Managers should implement measures to mitigate the risks posed by their AI-based systems • Managers must consider regulatory advice from other governmental bodies. • AI systems should involve algorithmic accountability to enable appropriate human direction and control		<i>[t]he use of AI in financial services will become more advanced, going beyond chatbots and credit scoring and fraud protection; consumers should be open-minded to its benefits – UK Manager.</i> <i>[m]y basic understanding of python as a programming language relates to how it helps me in having a conversation with the team; I think marketers need to acquire more of this technical knowledge, with even the management team needing to understand the basics of AI in business operations - – UK Manager</i> <i>[a]s part of my own professional development, I am doing a short course online to have a better understanding of AI. This is a computer course, but I need it. I am learning about blockchain, machine learning and deep learning, trying to make sense of it - Vietnam Manager</i>
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Appendix 1: Distinction between AI and financial services and the use of AI in marketing financial services